FEATURE 09 OF 14 | PRIORITY P0

False-Positive Handling

Priority	P0
Judging criteria served	Completeness (35) | Innovation (20)
Estimated effort	Medium (~3 hrs) (rough estimate)
Depends on	Rule-Based Detection, Risk Scoring

Purpose

A scam checker that cries wolf is worse than none: users stop trusting it. This feature makes sure ordinary messages - real bank alerts, OTP deliveries, delivery updates - are not marked High without enough evidence.

What it must do

- A message cannot reach High from a single weak signal.
- Safety advisories ("Never share your OTP") are not treated as requests.
- Links to known official domains are not marked suspicious.
- A test set of at least 20 normal messages and 20 scam messages is kept in the project and run automatically.
- Result wording for Low/Medium stays humble ("no strong warning signs", not "safe").

How it should work

- **Built-in guard by arithmetic:** the highest single rule is 40, below the High threshold of 60 - so High always needs at least two independent signals. State this explicitly in your pitch.
- **Negation filter:** ignore matches preceded by 'do not', 'never', 'don't' etc. (see feature 2).
- **Official-domain allowlist:** a small config list (e.g. sbi.co.in, hdfcbank.com, npci.org.in, and similar well-known domains you verify yourself). Links exactly matching these are not flagged.
- **Benign-pattern recognition:** typical transaction alerts (debited/credited, A/c XX1234, ref no.) lower confidence of threat rules; implement as a plain 'context note', not a hidden score reduction, so the maths remains visible.
- **Evaluation table:** run both test sets and print a confusion matrix (scams caught, normal messages wrongly flagged). Showing this in the demo is a strong credibility signal. Be honest: a small hand-made test set shows behaviour on those examples, not real-world accuracy.
- **User feedback hook (P1):** a 'This result seems wrong' button that stores the message ID and rule list locally for later review.

UI / UX notes

- Low result copy invites caution without promising safety.
- If a message contains an OTP but also a 'do not share' advisory, show a small chip: "Looks like a normal OTP message".

Edge cases and risks

- Real bank messages that contain a link: allowlisted domains only; anything else still gets flagged.
- A scammer copies the 'Never share OTP' sentence to look legitimate. Only the specific negated phrase is ignored; any real request elsewhere in the text still fires.

- Messages from friends containing 'urgent' or 'immediately': urgency alone stays Low.

Acceptance tests

Try this	Expected result
"Your OTP is 482913. Do not share it with anyone."	Low (0)
"Rs 500 debited from A/c XX1234. Never share OTP."	Low
"Urgent: meeting moved to 3pm"	Low (20 or less)
Scam copies 'Never share OTP' but also says 'send me the OTP'	OTP_REQUEST still fires
Run full 40-message test set	Confusion matrix printed; no normal message High

Demo tip for the judges

> Show one scam and one normal message side by side: High vs Low. Then show the test-set table.